Guard-Ian

› Pedestrian Safety Alert System

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CSCI 635 - Machine Learning
April 20, 2026

PROBLEM STATEMENT

Distracted walking, visual impairment and personal safety concerns leave millions of pedestrians exposed to unseen threats.

To address this, we developed a machine learning model that detects and classifies nearby objects such as vehicles, cyclists, and pedestrians.

Scope

This project focuses on image-based object classification as a foundational step toward a real-time pedestrian safety alert system.

The current scope is limited to:

- Developing and evaluating 3 ML models
- Classifying 5 object classes: bicycle, bus, car, person, and truck
- Training and evaluating on static images from the COCO dataset
- Comparing the results of the ML models for object classification

Dataset

COCO-2017 Dataset

- Contains 118k+ training images and 5k validation images with annotated real-world objects
- Ideal because it includes people, cars, trucks, bicycles, and buses in diverse road and urban scenes

Person

Car

Bicycle

Truck

Bus

Dataset

Bottleneck

Imbalanced Dataset

Person (Majority Class)

Bus (Minority)

Handling Class Imbalance

All three models addressed class imbalance differently, resulting in different levels of performance across frequent classes like people and rarer classes like bicycles and trucks.

Preprocessing Variance

Preprocessing techniques and data samples chosen were different for each of the three models.

Models

SVM

Support Vector Machine

SVM classifier trained on ResNet50 features extracted from COCO dataset.

CNN

Convolutional Neural Network

CNN trained from scratch with 4 convolutional and max pooling layers and 1 fully connected layer

YOLO

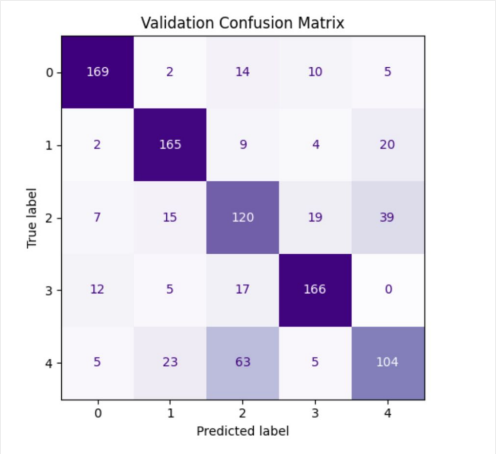
You Only Look Once

YOLOv8m pretrained CNN with 170 layers, 25.9M parameters, 25.9M gradients, at 512×512 resolution

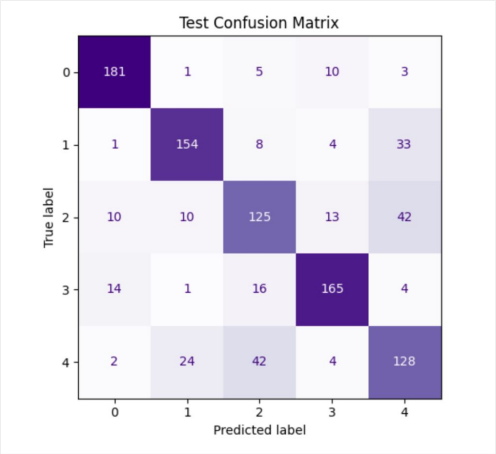
Support Vector Machine

- 1,400 samples/class = 7000 samples total
- Each image cropped to bounding boxes
- Features extracted using ResNet50 pre trained on ImageNet
  - ResNet50 Output = 2048-dim avg pool - **Bottleneck**
- LinearSVC classifier used with balanced class weights (C=1)
- Train / Validation / Test split: 70% / 15% / 15%
- Time taken: ~20 mins on GPU

val results	precision	recall	f1-score
bicycle	**0.87**	**0.84**	**0.86**
bus	**0.79**	**0.82**	**0.80**
car	0.54	0.60	0.57
person	**0.81**	**0.83**	**0.82**
truck	0.62	0.52	0.57



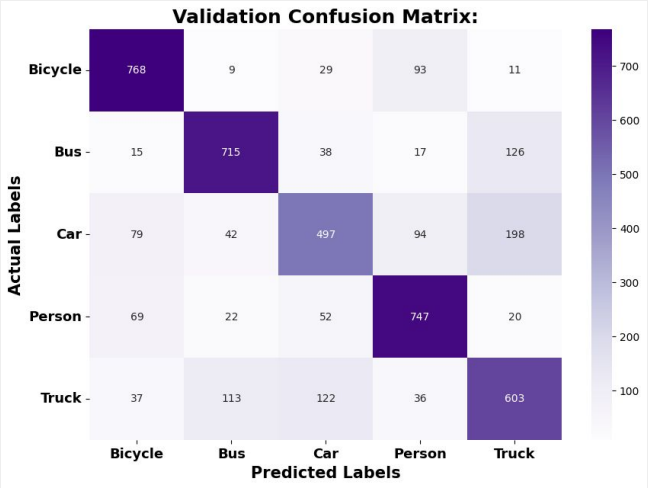
test results	precision	recall	f1-score
bicycle	**0.87**	**0.91**	**0.89**
bus	**0.81**	**0.77**	**0.79**
car	0.64	0.62	0.63
person	**0.84**	**0.82**	**0.83**
truck	0.61	0.64	0.62



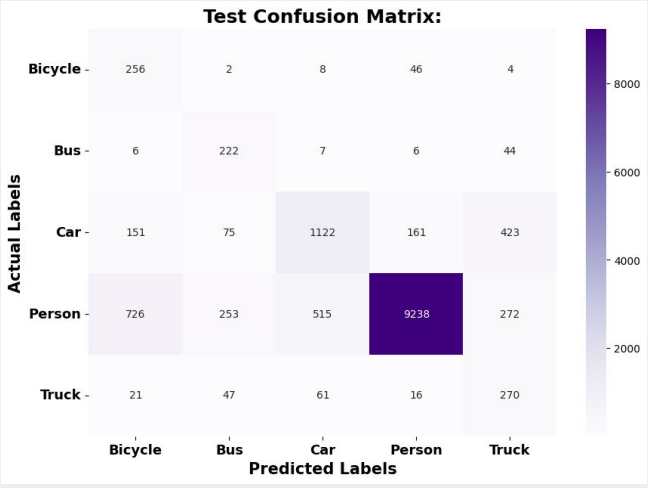
Convolutional Neural Network

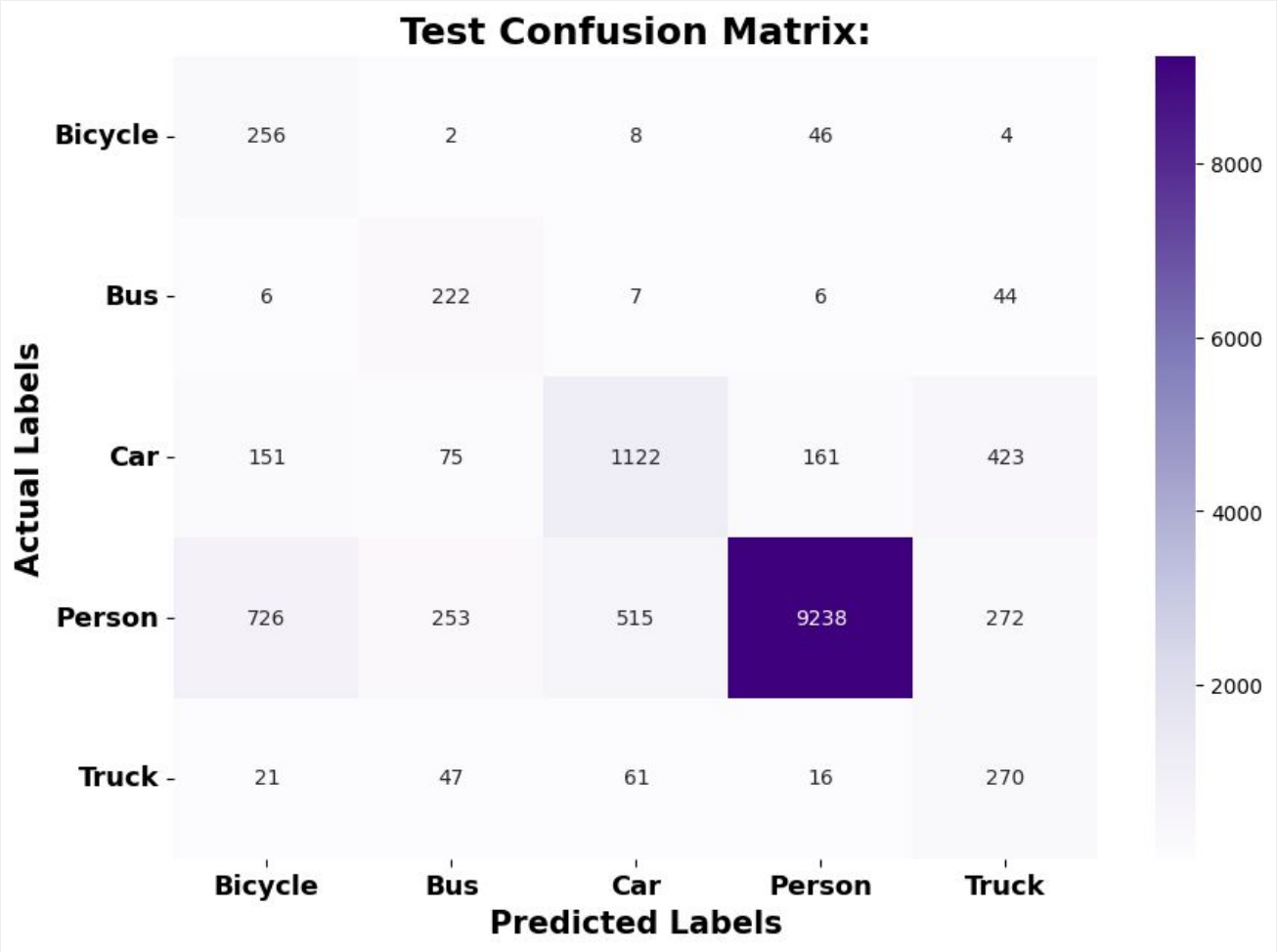
- 36,000 image samples for training evenly distributed. 6000 samples / class
- Each image cropped by bounding box
- Applied image augmentations to training set (color jitter, random flipping etc..)
- Features extracted from convolutional and max pooling layers
- 4 convolutional layers, 4 pooling layers. 1 fully connected layer
- 50% Dropout Rate
- Train/Validation split: 85%/15% stratified.
- Test set: 13K samples
- Time: 2 hours on GPU for 25 epochs

val results	precision	recall	f1-score	sample size
bicycle	**0.79**	**0.84**	**0.82**	**910**
bus	**0.79**	**0.78**	**0.79**	**911**
car	0.67	0.55	0.60	910
person	**0.76**	**0.82**	**0.79**	**910**
truck	0.63	0.66	0.65	911



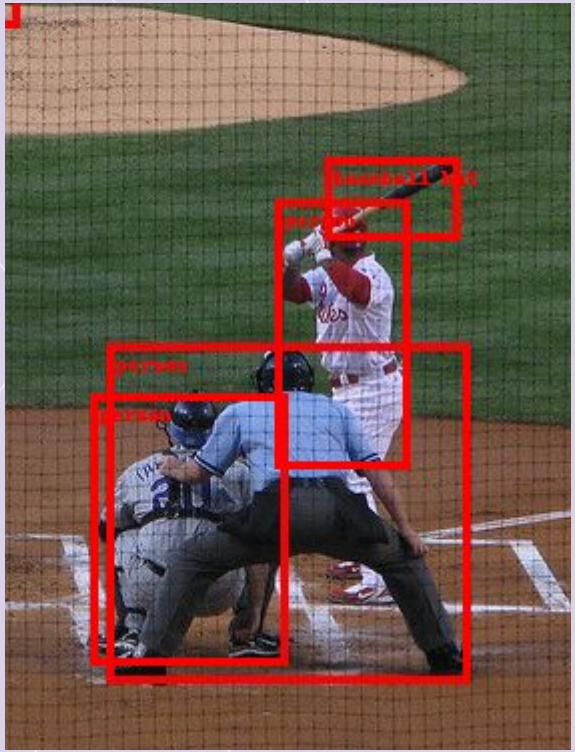
test results	precision	recall	f1-score	sample size
bicycle	0.22	**0.81**	0.35	316
bus	0.37	**0.78**	0.50	285
car	**0.65**	**0.58**	**0.62**	**1932**
person	**0.98**	**0.84**	**0.90**	**11004**
truck	0.27	**0.65**	0.38	415



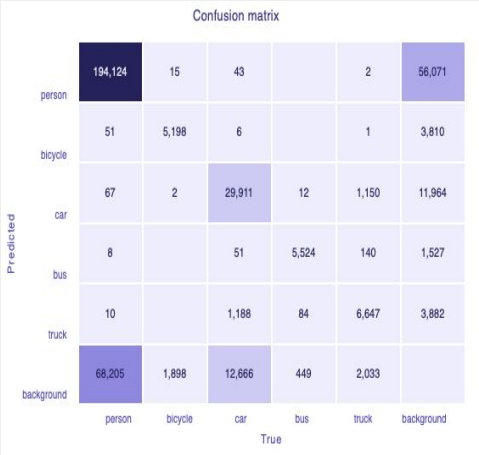

Test Confusion Matrix:
Actual Labels
Bicycle
Bus
Car
Person
Truck
256
2
8
46
4
6
222
7
6
44
151
75
1122
161
423
726
253
515
9238
272
21
47
61
16
270
Bicycle
Bus
Car
Person
Truck
Predicted Labels
8000
6000
4000
2000

YOLO V8m (Medium)

- 69,655 training samples, 2,948 test samples
- COCO annotations filtered and converted into YOLO format for 5 classes
- Custom weighted data loader integrated into YOLOv8m pipeline to oversample rare classes without modifying labels
- Model fine-tuned on prepared dataset with augmentations (mosaic, mixup, copy-paste) for robustness
- Performance analyzed using mAP, precision-recall curves, and confusion matrices on training and test sets
- Time taken: ~11 hours on GPU for 20 epochs



train results	precision	recall	mAP50	mAP50-95
bicycle	**0.73**	0.64	**0.70**	0.50
bus	0.86	**0.90**	**0.93**	**0.82**
car	0.83	**0.61**	0.73	**0.49**
person	**0.86**	0.66	0.79	0.55
truck	0.76	0.67	**0.76**	0.61



test results	precision	recall	mAP50	mAP50-95
bicycle	0.63	**0.51**	**0.56**	**0.34**
bus	0.76	**0.80**	**0.84**	**0.71**
car	0.75	0.59	0.67	0.45
person	**0.83**	0.67	0.77	0.53
truck	**0.58**	**0.51**	**0.57**	**0.42**

Best Model

	Precision	Recall	F1-Score	Accuracy
SVM	0.75	0.75	0.75	0.75
CNN	0.88	0.80	0.83	0.80
YOLO	0.71	0.62	-	0.68

Future Work

Expand Training Dataset

Acquire high-fidelity, first-person POV imagery to overcome current data scarcity and improve model robustness.

Advanced Object Tracking

Integrate tracking capabilities to calculate the relative approach speed of moving obstacles.

Real-Time Alert System

Develop a notification architecture that provides immediate visual or auditory cues upon obstacle detection.

Dynamic Danger Scoring

Implement an algorithmic scoring system to assess and rank the threat level of obstacles in the direct path.

What's on the market?

Glidance - Utilizes object detection to guide users via a handle and wheels

StellarTrek - GPS driven and Computer Vision. Identifies street addresses, entrances and traffic lights

Questions?

THANK YOU